



Federal Urdu University
of Arts, Science and Technology

ARTIFICIAL INTELLIGENCE
ASSIGNMENT # 01

MUNTAHA SHUMAIL

BSCS

SECTION: - C

SEMESTER: - 6TH

ROLL NO: - 23122167

SUBMITTED TO DR UZMA

Smart Parking Agent

1. Name of the Agent: -

Smart Parking Management Agent:

The Smart Parking Agent is an intelligent agent that helps drivers find available parking spaces. It uses sensors to detect empty and occupied parking spaces and takes actions to guide the driver toward a suitable parking space.

2. Agent Formulation: -

E. Environment:

- The environment is a parking area or parking garage containing:
- Parking Space.
- Vehicles.
- Entry and exit points.
- Paths inside the parking area.
- Other drivers.

S. Sensors:

The agent can use:

- Camera sensors to detect vehicles.
- Exit sensors to count vehicles.
- Location sensors to determine the vehicle's location.

A. Actuators:

The agent can use:

- Digital display boards show available spaces.
- App notifications to guide the driver.
- Automatic barriers at parking entrances and exits.

3. Environment Types: -

You can evaluate the Smart Parking environment using the common AI environment properties:

<u>Environment Type</u>		<u>Smart Parking Agent</u>
Fully Observable	Partially Observable	Partially Observable
Deterministic	Stochastic	Stochastic
Episodic	Sequential	Sequential
Static	Dynamic	Dynamic
Discrete	Continuous	Mostly Continuous
Single agent	Multi-agent	Multi-agent

Partially Observable: - The agent may not have complete information about every vehicle or parking space.

Stochastic: - Driver behavior and vehicle movement cannot always be predicted.

Sequential: - One action can affect future parking decisions.

Dynamic: - Vehicles continuously enter, leave, and move around the parking area.

Continuous: - Vehicles and traffic can move continuously.

Multi-agent: - Many drivers and vehicles interact with the parking system.

4. PEAS Framework: -

PEAS Component	Smart Parking Agent
P. Performance Measure	Find parking quickly, reduce waiting time, use parking spaces efficiently, reduce congestion
E. Environment	Parking lot/garage, vehicles, drivers, parking spaces, entry and exit roads
A. Actuators	Display boards, LED indicators, mobile notifications, parking barriers
S. Sensors	Cameras, ultrasonic sensors, entry sensors, location sensors

5. Graphical Representation: -

